

Signal transduction pathways link signal reception to response

The idea that plants are inert or passive is a common misconception. Nothing could be further from the truth. As shown in the examples of plant responses in **Figure 39.1**, plants must sense and integrate information about many facets of their environment. Although plant development is simpler than the development of multicellular animals, plant cells are just as complex as animal cells (see Figure 6.8). The molecular biology of plants is as complicated as that of animals: A Japanese plant (*Paris japonica*) currently has the largest genome on record, about fifty times greater than the human genome. Plants have among the largest genomes (see Table 21.1). At the levels of signal reception and signal transduction, your cells are not all that different from those of plants—the similarities far outweigh the differences. As an animal, however, your responses to environmental stimuli are generally quite different from those of plants. Animals commonly respond by movement; plants do so by altering growth and development.

As an example of a plant modifying its growth and development in response to environmental cues, consider a forgotten potato in the back corner of a kitchen cupboard. This modified underground stem, or tuber, has sprouted shoots from its “eyes” (axillary buds). These shoots, however, scarcely resemble those of a typical plant. Instead of sturdy stems and broad green leaves, this plant has ghostly pale stems and unexpanded leaves, as well as short, stubby roots (**Figure 39.2a**). These physical adaptations for growing in darkness, collectively referred to as **etiolation**, make sense if we consider that a young potato plant in nature usually encounters continuous darkness when sprouting underground. Under these circumstances, expanded leaves would be a hindrance to soil penetration and would be damaged as the shoots pushed through the soil. Because the leaves are unexpanded and underground, there is little evaporative loss of water and little requirement for an extensive root system to replace the water lost by transpiration. Moreover, the energy expended in producing green chlorophyll would be wasted because there is no light for photosynthesis. Instead, a potato plant growing in the dark allocates as much energy as possible to elongating its stems. This adaptation enables the shoots to break ground before the nutrient reserves in the tuber are exhausted. The etiolation response is one example of how a plant’s physical characteristics are tuned to its surroundings by complex interactions between environmental and internal signals.

When a shoot reaches light, the plant undergoes profound changes, collectively called **de-etiolation**

▼ **Figure 39.2** Light-induced de-etiolation (greening) of dark-grown potatoes.



- (a) **Before exposure to light.** A dark-grown potato has tall, spindly stems and nonexpanded leaves—morphological adaptations that enable the shoots to penetrate the soil. The roots are short, but there is little need for water absorption because little water is lost by the shoots.
- (b) **After a week's exposure to natural daylight.** The potato plant begins to resemble a typical plant with broad green leaves, short sturdy stems, and long roots. This transformation begins with the reception of light by a specific pigment, phytochrome.

(informally known as greening). Stem elongation slows; leaves expand; roots elongate; and the shoot produces chlorophyll. In short, it begins to resemble a typical plant (**Figure 39.2b**). In this section, we’ll examine this de-etiolation response as an example of how a plant cell’s reception of a signal—in this case, light—is transduced into a response (greening). Along the way, we will explore how studies of mutants provide insights into the molecular details of the stages of cell signal processing: reception, transduction, and response (**Figure 39.3**).

▼ **Figure 39.3** Review of a general model for signal transduction pathways. As discussed in Concept 11.1, a hormone or other kind of stimulus interacting with a specific receptor protein can trigger the sequential activation of relay proteins and also the production of second messengers that participate in the pathway. The signal is passed along, ultimately bringing about cellular responses. In this diagram, the receptor is on the surface of the target cell; in other cases, the stimulus interacts with receptors inside the cell.

